

assignment14 Life of stars I - EXTRA POINTS

 This is a preview of the draft version of the quiz

Started: Mar 23 at 7:05pm

Quiz Instructions

Use slides 9.1 small stars, white dwarf, supernovae



Question 1 10 pts

In a brown dwarf, gravity is balanced by pressure (slide)

☐

from radiation

☐

from electrons degeneracy

☐

from neutron degeneracy

☐

from heat



Question 2 10 pts

What is the upper limit to the mass of a white dwarf?

☐

about 1.4 solar masses

☐

about 2.8 solar masses

☐

about 1 solar mass

☐

There is no upper limit.

☐

There is an upper limit, but we do not yet know what it is.



Question 3 10 pts

What is true (slides)

☐

brown dwarf has masses proportional to their size

☐

planets have lithium in their atmosphere



brown dwarf has lithium in its atmosphere



brown dwarf sustain fusion for a long time



Question 4 10 pts

A star (no matter what its mass) spends most of its life:



as a planetary nebula.



as a T Tauri variable star.



as a red giant or supergiant.



as a protostar.



as a main sequence star.



Question 5 10 pts

In a white dwarf



there is no more fusion



There is still fusion to provide pressure against gravity



There is some fusion (deuterium to helium)



No idea



Question 6 10 pts

a deuterium is an isotope of



hydrogen



lithium



oxygen



helium



Question 7 10 pts

Our Sun is in _____ equilibrium



atmospherica



mechanical



barometric



hydrostatic



Question 8 10 pts

What is the ultimate fate of an isolated white dwarf?



As gravity overwhelms the electron degeneracy pressure, it will become a neutron star.



As gravity overwhelms the electron degeneracy pressure, it will explode as a supernova.



As gravity overwhelms the electron degeneracy pressure, it will explode as a nova.



The electron degeneracy pressure will eventually overwhelm gravity and the white dwarf will slowly evaporate.



It will cool down and become gradually dimmer.



Question 9 10 pts

What is a planetary nebula?



a planet surrounded by a glowing shell of gas



the ejected envelope ("cosmic burp") of a red giant surrounding a stellar core remnant



the disc of gas and dust surrounding a young star that will soon form a solar system



a type of young, medium mass star



the bipolar jets ejected by a T Tauri variable



Question 10 10 pts

When the Sun burn hydrogen to helium, the helium accumulates (like ashes)



helium will be sprad out all over



between the core and the outmost layer



in the core



in its outmost layer



Question 11 10 pts

When the Sun will become a white dwarf, its core will contain mostly (video)



deuterium



carbon



hydrogen



lithium



helium



Question 12 10 pts

in $E = m c^2$, c is



the speed of light and is a very big number



the number of nuclei being very fused and is a very big number



the number of nuclei being very fused and is a very small number



the speed of light and is a very small number



Question 13 10 pts

When a star's inward gravity and outward pressure are balanced, the star is said to be



in rotational equilibrium.



in hydrostatic equilibrium.



a stage 2 protostar.



in thermal expansion.



in gravitational collapse.



Question 14 10 pts

What inevitably forces a star like the Sun to evolve away from being a main sequence star?

☐

The carbon detonation explodes it as a type I supernova.

☐

The star uses up all its supply of hydrogen.

☐

The core loses all its neutrinos, so all fusion ceases.

☐

Helium builds up in the core, while the hydrogen burning shell expands.

☐

The core begins fusing iron.

**Question 15 10 pts**

who observed SN1572

(in 1572. use internet if not said in class, The supernova of 1572)

☐

Einstein

☐

Johannes Kepler

☐

Galileo

☐

Tycho Brahe

**Question 16 10 pts**

What is a white dwarf?

☐

the exposed core of a dead star, supported by neutron degeneracy pressure

☐

the name for the singularity at the center of a black hole

☐

the exposed core of a dead star, supported by electron degeneracy pressure

☐

a cool and very small main-sequence star with a mass of less than 1.4 solar masses

☐

a hot, but very small main-sequence star with a mass of less than 1.4 solar masses

**Question 17 10 pts**

(slide). The photons produced by fusion the core take about _____ years to reach the surface of the Sun

☐

10,000

☐

100,000

☐

1,000,000

☐

10



Question 18 10 pts

for a star 8-10 times the mass of the Sun, magnesium can be produced (slide)

☐

False

☐

magnesium is only synthesized in blue stars

☐

magnesium is only synthesized in supernovae

☐

True



Question 19 10 pts

In class, we talked about fusion.

during a fusion reaction, mass is lost and energy is released

☐

I don't know

☐

False

☐

True

Not saved

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